

KAILASH G

+91 9360431488 | gkailash.vs@gmail.com | Thanjavur, Tamil Nadu
linkedin.com/in/kailash-gurumoothy | codeberg.org/Kailash_Gurumoothi

SUMMARY

ECE Graduate with a strong foundation in Java, Object-Oriented Programming (OOPs), and system logic. Proven ability to solve algorithmic problems, troubleshoot hardware-software integration issues, and transition hardware logic into software applications. Seeking an Associate Software Engineer role to leverage analytical problem-solving skills and backend development knowledge.

TECHNICAL SKILLS

- **Programming:** Java (Core and OOP), Embedded C, SQL (Basic)
- **Systems & IoT:** Arduino, ESP32, Digital Electronics, GSM/GPS Modules, Sensors
- **Development Tools:** Eclipse, Arduino IDE, Git, LTSpice, Tanner EDA
- **Core Concepts:** Data Structures & Algorithms, System Design Basics, SDLC
- **Soft Skills:** Technical Troubleshooting, Documentation, Team Collaboration

PROJECT EXPERIENCE

Accident Detection and Instant Emergency Alert System (IoT)

2024 – 2025

Role: System Designer & Developer

- Designed an autonomous safety system capable of detecting vehicle accidents in real-time using acceleration sensors and vibration data.
- Programmed the microcontroller (Arduino) logic to process sensor inputs and filter false positives, ensuring high system reliability.
- Integrated GPS and GSM modules to trigger instant location-based alerts to emergency contacts within 10 seconds of incident detection.
- Conducted rigorous hardware testing to verify signal integrity between the sensor nodes and the communication modules.
- **Tech Stack:** Embedded C, Arduino, GPS/GSM, IoT Sensors.

Low Power Design of TSDC D-Flip Flop (VLSI)

2023 – 2024

Role: Circuit Design Engineer

- Engineered a high-efficiency digital circuit (D-Flip Flop) using CMOS technology nodes (90nm, 65nm, 45nm).
- Implemented MTCMOS (Multi-Threshold CMOS) techniques to reduce leakage current, achieving a 28% reduction in overall power consumption.
- Optimized the design for propagation delay, balancing speed and energy efficiency in simulation environments.
- **Tech Stack:** Tanner EDA, LTSpice, Digital Logic Design.

EDUCATION

SASTRA Deemed University, Thanjavur

2021 – 2025

B.Tech in Electronics and Communication Engineering | CGPA: 6.39/10

Sri Mahadeva Guruji Matric Higher Secondary School

2020 – 2021

Higher Secondary Certificate (HSC) | Percentage: 84.1%

Sri Mahadeva Guruji Matric Higher Secondary School

2018 – 2019

Secondary School Leaving Certificate (SSLC) | Percentage: 80.4%

ADDITIONAL INFORMATION

- **Learning:** Currently upskilling in Spring Boot and Advanced Data Structures to enhance backend engineering capabilities.
- **Interests:** Building DIY electronics projects, reverse engineering consumer gadgets, and solving algorithmic problems on Leetcode.